Predicting T-cell receptor– epitope specificity from sequence

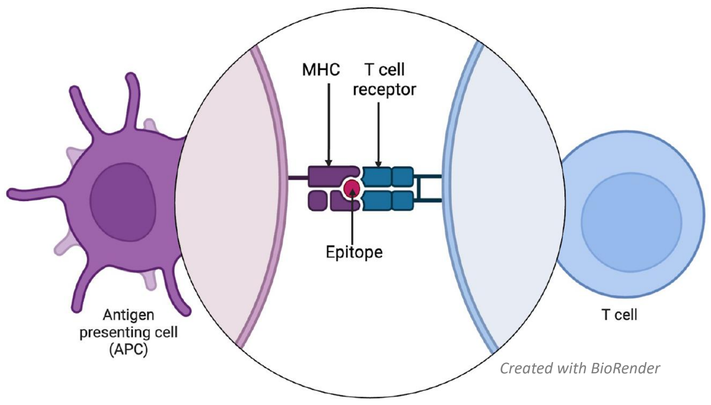
Olga Lyudovyk, Artem Streltsov, Yuval Elhanati, Quaid Morris, Benjamin Greenbaum

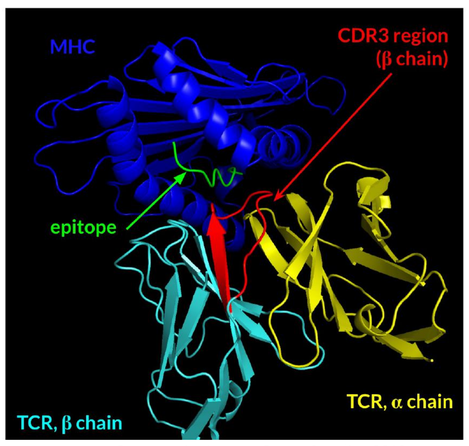
Weill Cornell Medicine

Memorial Sloan Kettering Cancer Center

INTRODUCTION

T-cell specificity – understanding which T-cells recognize which antigens – is an unsolved problem with potential to improve patient care in infectious diseases, auto-immune diseases and cancer. It is challenging due to high variability of both T-cell receptors (TCRs) and epitopes they recognize and due to lack of experimental data.

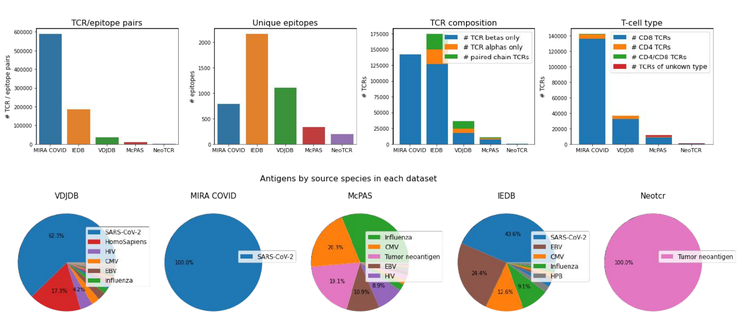




Created with ImmuneScape, Songling 2019

DATASETS

Current public datasets consist of binder pairs of epitopes of predominantly viral origin and sequences of TCR beta chains. Paired alpha/beta chain sequences are much less frequent since obtaining them requires costly single cell sequencing. We leveraged MIRA, VDJB and McPAS for training IEDB and NeoTCR for validation.

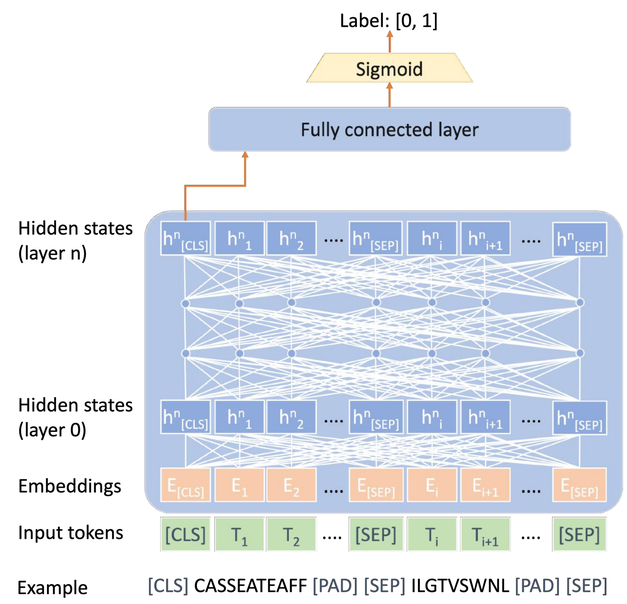


BERTie in a nutshell

- Accurately predicting TCR specificity could have wide practical applications for many disease areas including cancer immunotherapy and cell therapies
- We explored the ability of a sequence-based transformer model to predict TCR-epitope specificity
- Relevant for clinical applications, BERTie requires only CDR3 of TCR beta chains to predict TCR specificity
- It performs well for unseen epitopes similar in sequence to epitopes in training data
- However, neither BERTie nor any other evaluated sequence models generalize to cancer setting
- Cancer neoantigen-specific experimental datasets and potentially approaches beyond sequence are required

MODEL

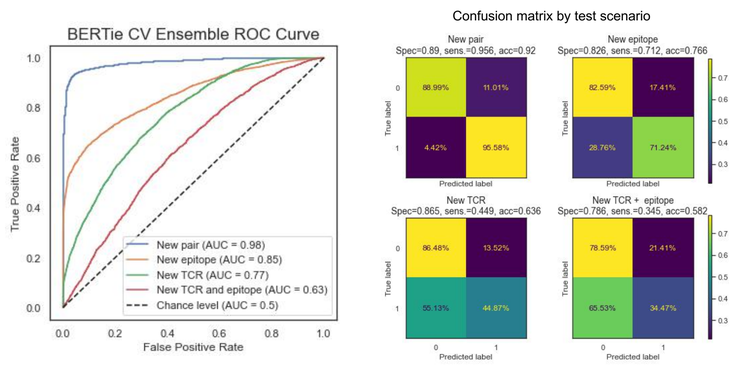
BERTie (BERt for TCRs and Immunogenic Epitopes) is a BERT model coupled with a fully connected layer trained to classify TCR-epitope pairs as cognate or non-cognate.



BERTie is trained with binary cross entropy loss with or without explicit training of the masked language model (MLM). We performed ablations for different encodings, with 5-fold cross-validation ensemble of 3-gram based models performing the best, and MLM task not contributing to significant performance improvement.

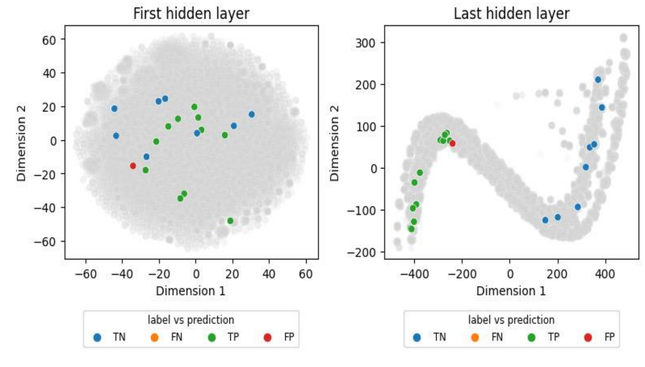
RESULTS

BERTie was evaluated on 4 scenarios of increasing difficulty: novel pairs of TCRs and epitopes seen during training; an unseen TCR paired with a seen epitope; an unseen epitope paired with a seen TCR; a pair of both TCR and epitope not seen during training. Even in the last challenging scenario, BERTie can leverage sequence similarity to seen TCRs and epitopes to predict cognate pairs better than chance.



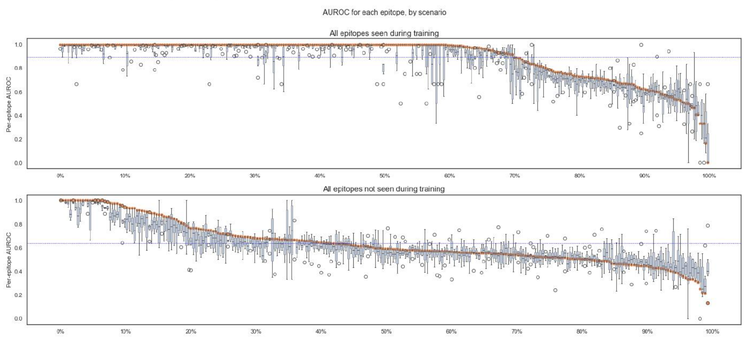
TCRs specific for epitope **SMWSFNPNET** exhibit a motif with conserved residues potentially important for epitope recognition.

Exploring weights of BERTie's first and last hidden layers shows progressive differentiation between specific and non-specific TCRs for epitope **SMWSFNPNET**.

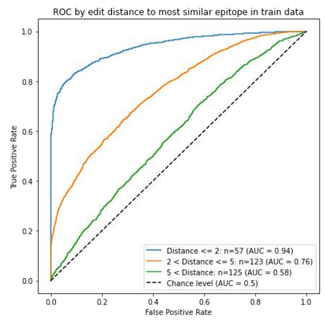


INTERPRETING RESULTS

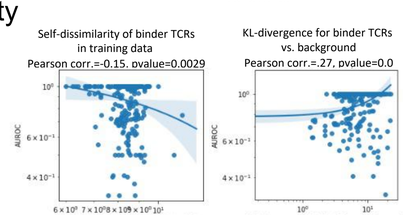
BERTie accurately predicts novel TCRs for epitopes it has seen during training. However, per-epitope performance for unseen epitopes is lower.



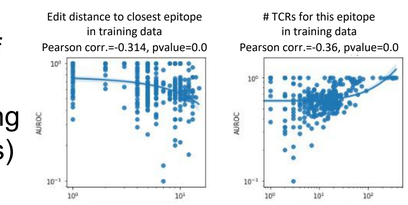
Performance for unseen epitopes whose sequences are similar to epitopes seen during training is significantly better than for unseen epitopes with very different sequences, as could be expected.



For seen epitopes, similarity of binder TCRs to each other and their difference from other TCRs correlate with good performance.

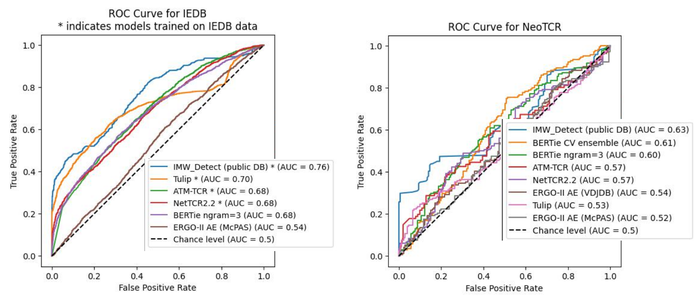


For unseen epitopes, distance to closest seen epitope and the number of TCRs specific for this epitope seen during training (paired with other epitopes) explain predictability.



EVALUATION ON NEW DATA

BERTie performs on new data (IEDB) similarly to models whose training data included this dataset and models trained on paired alpha/beta chains. However, all models fail to generalize to cancer neoantigen-derived epitopes (NeoTCR), likely because these epitopes are very different in sequence from largely viral training data, thus sequence similarity that models rely on is not informative.



CONCLUSIONS

- Sequence-based BERTie model has high accuracy for predicting new TCRs for the epitopes for which training data exist and generalizes well to epitopes similar to known epitopes.
- Performance on cancer neoantigen-specific TCRs is very poor by all methods, likely because these neoantigens are very different from largely viral antigens in the training data.
- Since it is impractical to obtain experimental binding data for all possible neoantigens, relying on sequence similarity might not be sufficient to predict TCR specificities for cancer neoantigens..

REFERENCES

Datasets: Nolan 2020, Bagayev 2019, Tickotsky 2017, Zhou 2024, Vita 2018

BERT model: Devlin 2018, huggingface.co

Benchmarked models: Springer 2021, Cai 2022, ImmuneWatch Detect V.1.0 2024, Jensen 2024, Meynard-Piganeau 2024.